

E4 Trigonometric functions A level only

Name: _____

Class: _____

Date: _____

Time: **249 minutes**

Marks: **204 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

$$p(x) = 30x^3 - 7x^2 - 7x + 2$$

- (a) Prove that $(2x + 1)$ is a factor of $p(x)$

(2)

- (b) Factorise $p(x)$ completely.

(3)

- (c) Prove that there are no real solutions to the equation

$$\frac{30 \sec^2 x + 2 \cos x}{7} = \sec x + 1$$

(5)

(Total 10 marks)

Q2.

Prove the identity $\cot^2 \theta - \cos^2 \theta \equiv \cot^2 \theta \cos^2 \theta$

(Total 3 marks)

Q3.

A curve has equation $y = 2x \cos 3x + (3x^2 - 4) \sin 3x$

- (a) Find $\frac{dy}{dx}$, giving your answer in the form $(mx^2 + n) \cos 3x$, where m and n are integers.

(4)

- (b) Show that the x -coordinates of the points of inflection of the curve satisfy the equation

$$\cot 3x = \frac{9x^2 - 10}{6x}$$

(4)

(Total 8 marks)

Q4.

- (a) Find $\frac{dy}{dx}$ when

$$y = e^{3x} + \ln x$$

(2)

- (b) (i) Given that $u = \frac{\sin x}{1 + \cos x}$, show that $\frac{du}{dx} = \frac{1}{1 + \cos x}$.

(3)

(ii) Hence show that if $y = \ln \left(\frac{\sin x}{1 + \cos x} \right)$, then $\frac{dy}{dx} = \operatorname{cosec} x$.

(2)

(Total 7 marks)

Q5.

(a) Show that

$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)}$$

can be written as $\operatorname{cosec}^2 x$.

(3)

(b) Hence solve the equation

$$\frac{\sec^2 x}{(\sec x + 1)(\sec x - 1)} = \operatorname{cosec} x + 3$$

giving the values of x to the nearest degree in the interval $-180^\circ < x < 180^\circ$.

(6)

(c) Hence solve the equation

$$\frac{\sec^2(2\theta - 60^\circ)}{(\sec(2\theta - 60^\circ) + 1)(\sec(2\theta - 60^\circ) - 1)} = \operatorname{cosec}(2\theta - 60^\circ) + 3$$

giving the values of θ to the nearest degree in the interval $0^\circ < \theta < 90^\circ$.

(2)

(Total 11 marks)

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Q6.

(a) Express $3 \cos x + 2 \sin x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$, giving your value of α to the nearest 0.1° .

(3)

(ii) Hence find the minimum value of $3 \cos x + 2 \sin x$ and the value of x in the interval

$0^\circ < x < 360^\circ$ where the minimum occurs. Give your value of x to the nearest 0.1° .

(3)

(b) (i) Show that $\cot x - \sin 2x = \cot x \cos 2x$ for $0^\circ < x < 180^\circ$.

(3)

- (ii) Hence, or otherwise, solve the equation

$$\cot x - \sin 2x = 0$$

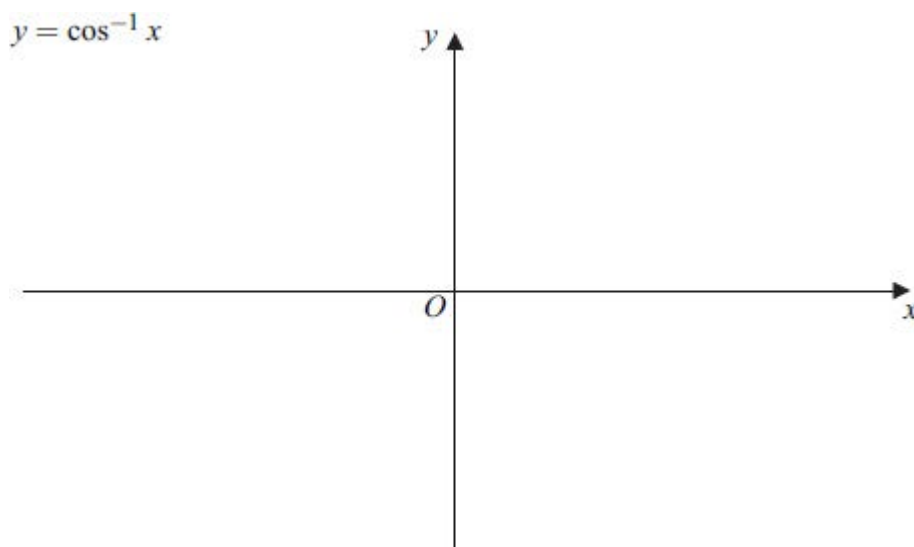
in the interval $0^\circ < x < 180^\circ$.

(3)

(Total 12 marks)

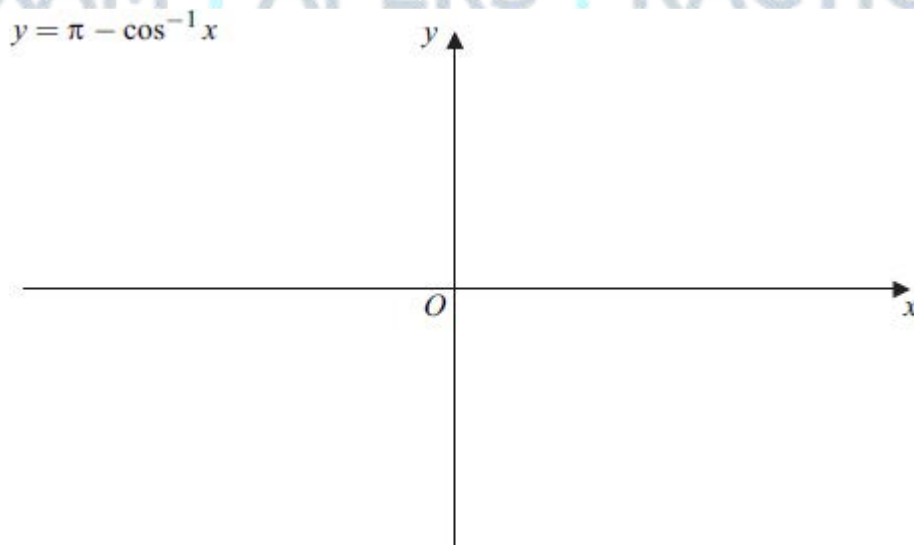
Q7.

- (a) Sketch the graph of $y = \cos^{-1} x$, where y is in radians. State the coordinates of the end points of the graph.



(2)

- (b) Sketch the graph of $y = \pi - \cos^{-1} x$, where y is in radians. State the coordinates of the end points of the graph.



(2)

(Total 4 marks)

Q8.

- (a) Show that the equation

$$\frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta} = 32$$

can be written in the form

$$\operatorname{cosec}^2 \theta = 16$$

(4)

- (b) Hence, or otherwise, solve the equation

$$\frac{1}{1 + \cos(2x - 0.6)} + \frac{1}{1 - \cos(2x - 0.6)} = 32$$

giving all values of x in radians to two decimal places in the interval $0 < x < \pi$.

(5)

(Total 9 marks)

Q9.

- (a) Given that $x = \frac{\sin y}{\cos y}$, use the quotient rule to show that

$$\frac{dx}{dy} = \sec^2 y$$

(3)

- (b) Given that $\tan y = x - 1$, use a trigonometrical identity to show that

$$\sec^2 y = x^2 - 2x + 2$$

(2)

- (c) Show that, if $y = \tan^{-1}(x - 1)$, then

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

(1)

- (d) A curve has equation $y = \tan^{-1}(x - 1) - \ln x$.

- (i) Find the value of the x -coordinate of each of the stationary points of the curve.

(4)

- (ii) Find $\frac{d^2 y}{dx^2}$.

(2)

(iii) Hence show that the curve has a minimum point which lies on the x -axis.

(2)

(Total 14 marks)

Q10.

(a) Given that $x = \tan(3y + 1)$:

(i) find $\frac{dx}{dy}$ in terms of y ;

(2)

(ii) find the value of $\frac{dy}{dx}$ when $y = -\frac{1}{3}$.

(2)

(b) Sketch the graph of $y = \tan^{-1} x$.

(2)

(Total 6 marks)

Q11.

(a) Solve the equation $\sec x = -5$, giving all values of x in radians to two decimal places in the interval $0 < x < 2\pi$.

(3)

(b) Show that the equation

$$\frac{\operatorname{cosec} x}{1 + \operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1 - \operatorname{cosec} x} = 50$$

can be written in the form

$$\sec^2 x = 25$$

(4)

(c) Hence, or otherwise, solve the equation

$$\frac{\operatorname{cosec} x}{1 + \operatorname{cosec} x} - \frac{\operatorname{cosec} x}{1 - \operatorname{cosec} x} = 50$$

giving all values of x in radians to two decimal places in the interval $0 < x < 2\pi$.

(3)

(Total 10 marks)

Q12.

(a) (i) Solve the equation $\operatorname{cosec} \theta = -4$ for $0^\circ < \theta < 360^\circ$, giving your answers to the

nearest 0.1° .

(2)

- (ii) Solve the equation

$$2 \cot^2(2x + 30^\circ) = 2 - 7 \operatorname{cosec}(2x + 30^\circ)$$

for $0^\circ < x < 180^\circ$, giving your answers to the nearest 0.1° .

(6)

- (b) Describe a sequence of two geometrical transformations that maps the graph of $y = \operatorname{cosec} x$ onto the graph of $y = \operatorname{cosec}(2x + 30^\circ)$.

(4)

(Total 12 marks)

Q13.

- (a) (i) Sketch the graph of $y = \sin^{-1} x$, where y is in radians. State the coordinates of the end points of the graph.

(3)

- (ii) By drawing a suitable straight line on your sketch, show that the equation

$$\sin^{-1} x = \frac{1}{4}x + 1$$

has only one solution.

(2)

- (b) The root of the equation $\sin^{-1} x = \frac{1}{4}x + 1$ is α . Show that $0.5 < \alpha < 1$.

(2)

- (c) The equation $\sin^{-1} x = \frac{1}{4}x + 1$ can be rewritten as $x = \sin\left(\frac{1}{4}x + 1\right)$.

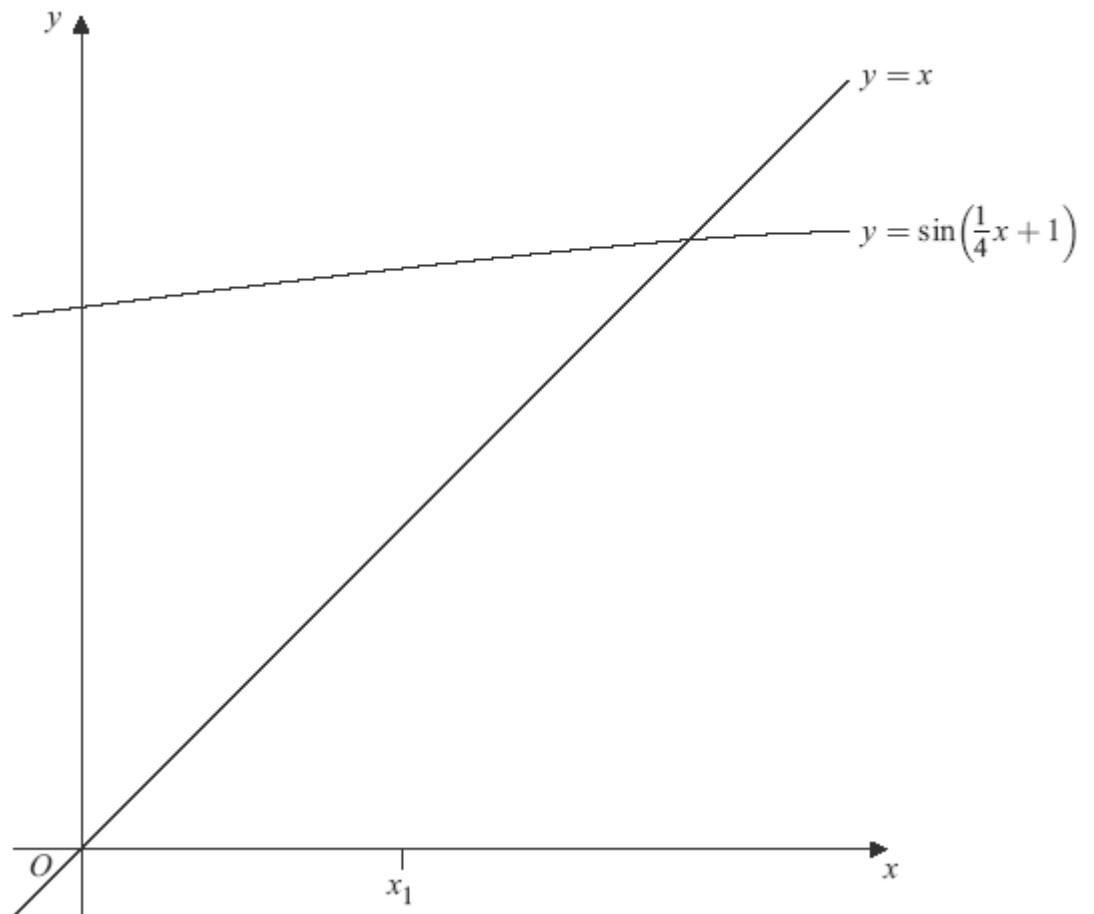
- (i) Use the iteration $x_{n+1} = \sin\left(\frac{1}{4}x_n + 1\right)$ with $x_1 = 0.5$ to find the values of x_2 and x_3 , giving your answers to three decimal places.

(2)

- (ii) The sketch on the figure below shows parts of the graphs of

$$y = \sin\left(\frac{1}{4}x + 1\right) \quad \text{and} \quad y = x, \quad \text{and the position of } x_1.$$

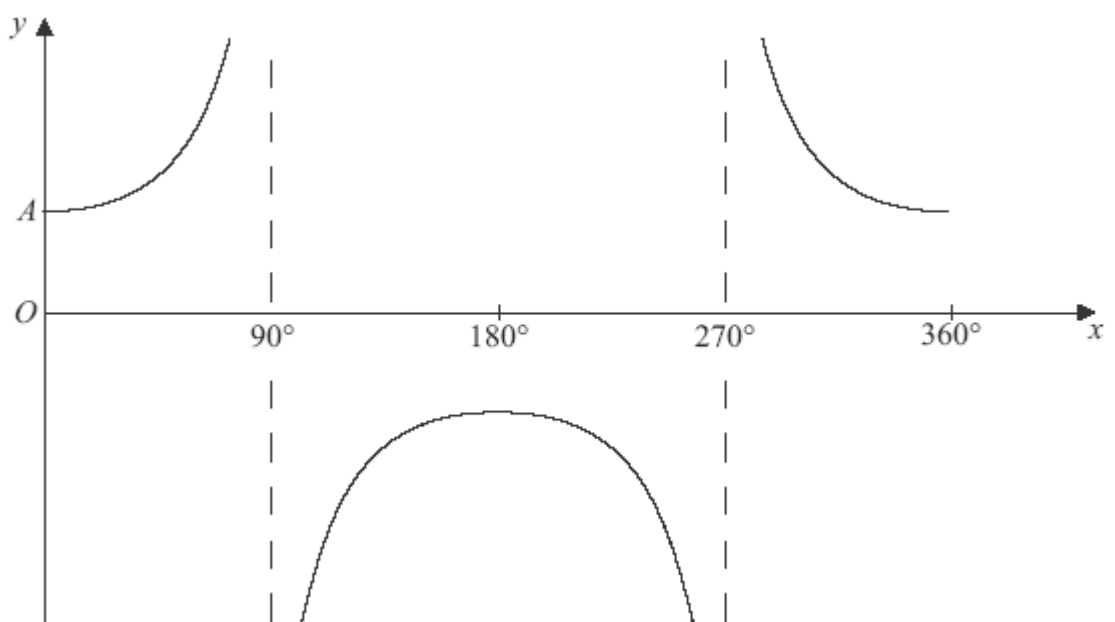
On the figure below, draw a cobweb or staircase diagram to show how convergence takes place, indicating the positions of x_2 and x_3 on the x -axis.



(2)
(Total 11 marks)

Q14.

- (a) The diagram shows the graph of $y = \sec x$ for $0^\circ \leq x \leq 360^\circ$.



The point A on the curve is where $x = 0$. State the y -coordinate of A .

(1)

- (b) Solve the equation $\sec x = 2$, giving all values of x in degrees in the interval $0^\circ \leq x \leq 360^\circ$.

(2)

(Total 3 marks)

Q15.

- (a) Solve the equation $\sec x = \frac{3}{2}$, giving all values of x to the nearest degree in the interval $0^\circ < x < 360^\circ$.

(2)

- (b) By using a suitable trigonometrical identity, solve the equation

$$2 \tan^2 x = 10 - 5 \sec x$$

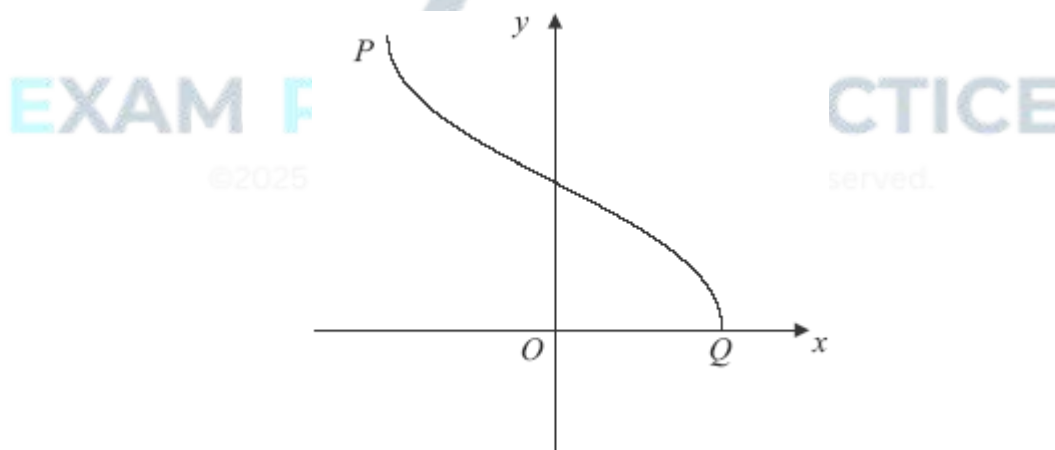
giving all values of x to the nearest degree in the interval $0^\circ < x < 360^\circ$.

(6)

(Total 8 marks)

Q16.

The sketch shows the graph of $y = \cos^{-1} x$.



- (a) Write down the coordinates of P and Q , the end points of the graph.
- (b) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos^{-1} x$ onto the graph of $y = 2 \cos^{-1}(x - 1)$.
- (c) Sketch the graph of $y = 2 \cos^{-1}(x - 1)$.

(2)

(4)

(2)

- (d) (i) Write the equation $y = 2 \cos^{-1}(x - 1)$ in the form $x = f(y)$.

(2)

- (ii) Hence find the value of $\frac{dx}{dy}$ when $y = 2$.

(3)

(Total 13 marks)

Q17.

- (a) Solve the equation $\cot x = 2$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(2)

- (b) Show that the equation $\operatorname{cosec}^2 x = \frac{3 \cot x + 4}{2}$ can be written as

$$2 \cot^2 x - 3 \cot x - 2 = 0$$

(2)

- (c) Solve the equation $\operatorname{cosec}^2 x = \frac{3 \cot x + 4}{2}$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q18.

- (a) Solve the equation $\sec x = 3$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(3)

- (b) Show that the equation $\tan^2 x = 2 \sec x + 2$ can be written as $\sec^2 x - 2 \sec x - 3 = 0$.

(2)

- (c) Solve the equation $\tan^2 x = 2 \sec x + 2$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(4)

(Total 9 marks)

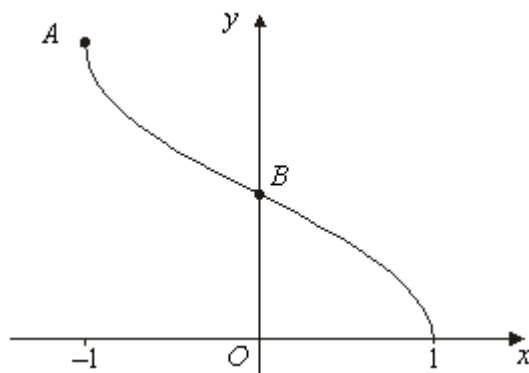
Q19.

Describe a sequence of **two** geometrical transformations that maps the graph of $y = \sec x$ onto the graph of $y = 1 + \sec 3x$.

(Total 4 marks)

Q20.

The diagram shows the curve $y = \cos^{-1}x$ for $-1 \leq x \leq 1$

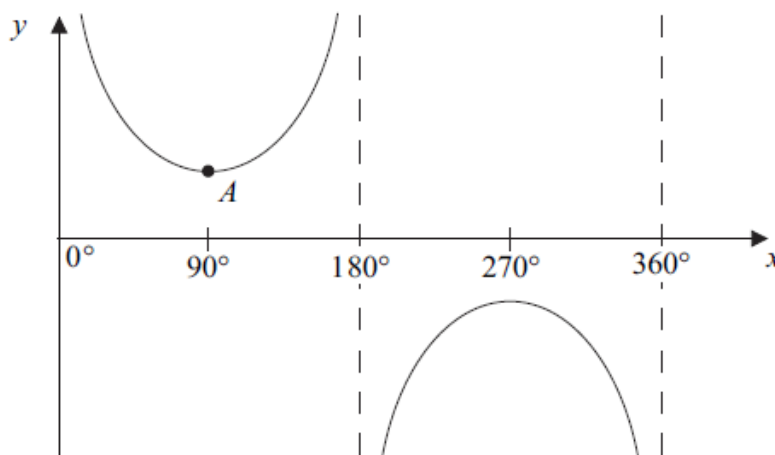


- (a) Write down the exact coordinates of the points A and B . (2)
- (b) The equation $\cos^{-1}x = 3x + 1$ has only one root. Given that the root of this equation is α , show that $0.1 \leq \alpha \leq 0.2$. (2)
- (c) Use the iteration $x_{n+1} = \frac{1}{3}(\cos^{-1}x_n - 1)$ with $x_1 = 0.1$ to find the values of x_2 , x_3 and x_4 , giving your answers to three decimal places. (3)

(Total 7 marks)

Q21.

- (a) Solve the equation $\operatorname{cosec} x = 2$, giving all values of x in the interval $0^\circ < x < 360^\circ$. (2)
- (b) The diagram shows the graph of $y = \operatorname{cosec} x$ for $0^\circ < x < 360^\circ$.



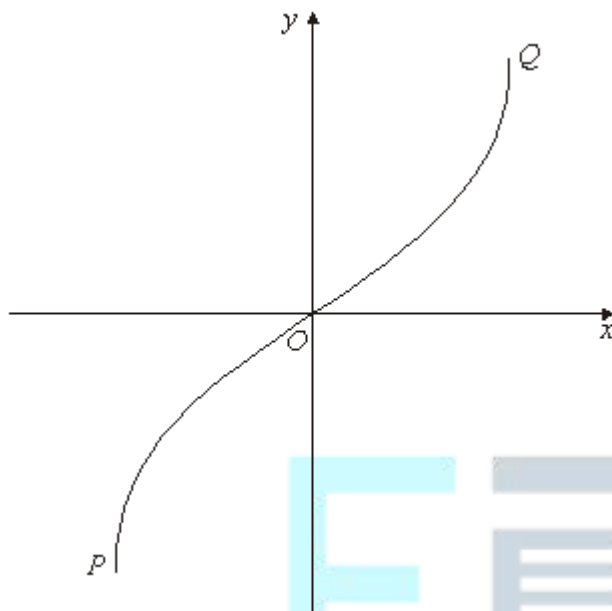
Sketch the graph of $y = |\operatorname{cosec} x|$ for $0^\circ < x < 360^\circ$.

(2)

(Total 4 marks)

Q22.

- (a) The sketch shows the graph of $y = \sin^{-1} x$.



Write down the coordinates of the points P and Q , the end-points of the graph.

(2)

- (b) Sketch the graph of $y = -\sin^{-1}(x - 1)$.

(3)

(Total 5 marks)

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Q23.

- (a) Solve the equation $\sec x = 5$, giving all the values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(3)

- (b) Show that the equation $\tan^2 x = 3 \sec x + 9$ can be written as

$$\sec^2 x - 3 \sec x - 10 = 0$$

(2)

- (c) Solve the equation $\tan^2 x = 3 \sec x + 9$, giving all the values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 9 marks)

Q24.

It is given that $\tan^2 x = \sec x + 11$.

- (a) Show that the equation $\tan^2 x = \sec x + 11$ can be written in the form

$$\sec^2 x - \sec x - 12 = 0$$

(2)

- (b) Hence show that $\cos x = \frac{1}{4}$ or $\cos x = -\frac{1}{3}$.

(3)

- (c) Hence, or otherwise, solve the equation $\tan^2 x = \sec x + 11$, giving all values of x to the nearest degree in the interval $0^\circ < x < 360^\circ$.

(3)

(Total 8 marks)

Q25.

- (a) Sketch the graph of $y = \tan^{-1} x$.

(2)

- (b) (i) By drawing a suitable straight line on your sketch, show that the equation $\tan^{-1} x = 2x - 1$ has only one root.

(2)

- (ii) Given that the root of this equation is α , show that $0.8 < \alpha < 0.9$.

(2)

- (c) Use the iteration $x_{n+1} = \frac{1}{2} (\tan^{-1} x_n + 1)$ with $x_1 = 0.8$ to find the value of x_3 , giving your answer to two significant figures.

(3)

(Total 9 marks)